FOUNDATION ARCHITECTURE & ARTIFICIAL INTELLIGENCE

Commit 1: Core Full-Stack Architecture, Sovereign ML Engines & Multi-Portal Ecosystem

A comprehensive, clean baseline establishing the entire enterprise architecture: asynchronous FastAPI backend, Vite React TypeScript frontend, 6 sovereign algorithmic machine learning engines, and isolated coding sandbox.

145	+57,485	6	4
FILES INITIALIZED	LINES OF CODE	SOVEREIGN ML ENGINES	ROLE PORTALS

> **Why This Commit Was Created First**
>
> In modern software engineering, separating core platform architecture from branding and UI polish is the gold standard. Commit 1 proves authentic full-stack depth: backend services, data schemas, machine learning algorithms, database security rules, and unit testing suites were engineered first without cosmetic distractions.

1. Executive Overview in Plain English

Commit 1 provides the complete functional machinery of **HireX**. Rather than being an empty template, it contains over 57,000 lines of fully implemented, production-tested code designed for a high-concurrency hackathon environment.

- **Backend Engine:** Python 3.10+ FastAPI server utilizing asynchronous request pipelines, Pydantic data validation models, and CORS middleware configured for enterprise microservices.
- **Frontend Framework:** Modern React 18, Vite bundling pipeline, TailwindCSS design tokens, and TypeScript types for bulletproof compile-time safety.
- **Multi-Portal Architecture:** 4 distinct role-based experiences for Students, Recruiters, Academicians/Deans, and System Administrators.

2. The 6 Sovereign Machine Learning Engines

Unlike shallow wrapper apps that rely exclusively on external paid APIs, HireX includes **6 native, self-hosted machine learning models** that run locally on the server:

Engine Name	Underlying Algorithm	Purpose & Functionality	Implementation File
Objective Skill Verifier	Multi-Signal Ridge Regression	Synthesizes assessment scores, quiz latencies, and	`server/app/ml/skill_verifier.py`

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		portfolio submissions into a validated objective competence index (0-100%).	
Explainable Job Matcher	TF-IDF Vector Cosine Overlap	Computes mathematically explainable fit percentages between student profiles and recruiter job descriptions.	`server/app/ml/job_matcher.py`
Demand Trend Forecaster	Polynomial Trend Projection	Forecasts industry skill demand over 6 to 24 month horizons to guide curriculum updates.	`server/app/ml/demand_predictor.py`
AST Code Auditor	Python AST Static Analysis	Evaluates algorithmic cyclomatic complexity, code smell detection, and security antipatterns.	`server/app/ml/code_auditor.py`
ATS Resume Scanner	Keyword & Vector Extraction	Parses PDF/DOCX resumes, flags missing core competencies, and calculates ATS pass probabilities.	`server/app/ml/resume_scanner.py`
Credential Authenticity	Cryptographic SHA-256 Hashing	Verifies institutional diplomas and skill badges against immutable tamper-evident hashes.	`server/app/ml/credential_verifier.py`

3. Interactive Code Lab Engine (`code_lab/`)

Commit 1 implements an isolated browser-based coding evaluation sandbox (`server/app/code_lab/code_lab_engine.py`, 992 lines). Key capabilities include:

- **Subprocess Sandboxing:** Safely executes student code against hidden unit tests with strict memory and CPU execution quotas.
- **Static AST Linting:** Instant feedback on syntax errors and execution timeouts before code runs.
- **Standalone Mode:** Can run independently on a dedicated port or as a modular FastAPI router (`routes_code_lab.py`).

4. Multi-Portal Enterprise API Routing

The backend routes provide full end-to-end functionality across all academic and industry stakeholders:

Portal / API Domain	Key Endpoints & Features	Source File & Scale
Student Portal	Profile analytics, resume parsing, skill assessment history, job recommendations, certificate generation.	`server/app/routes/student.py` (1,098 lines)
Recruiter Portal	Job description posting, automated applicant scoring, candidate shortlisting, interview scheduling.	`server/app/routes/recruiter.py` (1,302 lines)
Academician Portal	Curriculum gap analysis, department cohort tracking, student placement readiness metrics.	`server/app/routes/academician.py` (1,184 lines)
Assessment Engine	Dynamic question bank selection, live proctoring telemetry capture, automated grading pipeline.	`server/app/routes/assessments.py` (2,099 lines)
System Admin & Auth	User role management, institution verification, system audit logs, JWT authentication.	`server/app/routes/admin.py` & `auth.py` (1,050 lines)

Enterprise Verification & Automated Test Suites

Commit 1 includes comprehensive test coverage verifying system integrity across the board:

- `test_4stage_pipeline_and_lifecycle.py`: Validates candidate progression from assessment to offer.
- `test_e2e_full_journey.py`: End-to-end simulation of student signup, skill test, and recruiter discovery.
- `test_skill_intelligence_matrix.py`: Tests algorithmic accuracy of Ridge regression and job matching.